

ABOUT

M.S. in CS (Digital Skills for Health Transformation track) at **Institut Polytechnique de Paris (IP Paris)**; B.S. in Data Science from **SUSTech**; Research Assistant at **SUSTech-NLP** (adv. Prof. Guanhua Chen). I work on **how heterogeneous-capability LLMs should collaborate**: in **RouterXBench** I built a 3-axis evaluator and identified *internal hidden states* as a more discriminative routing signal than verbose, logit, and external-embedding alternatives; in **Scope-then-Cover** I exploit a capability asymmetry inside exploration: a strong actor scopes a small high-recall subspace while frozen cheap workers cover it in parallel, exposed as a single `explore` action.

EDUCATION

✳ Institut Polytechnique de Paris

M.S. in Computer Science (Digital Skills for Health Transformation track)

Paris, France

Sep 2025 – Present

- **ReflexSQL**: an exploration-and-memory Text-to-SQL agent on Spider 2.0-Lite [Poster].

🔥 Southern University of Science and Technology (SUSTech)

B.S. in Data Science and Big Data Technology

Shenzhen, China

Sep 2021 – Jul 2025

- Bachelor's thesis on LLM-SLM collaborative routing (continued in **RouterXBench**).

PUBLICATIONS

Towards Fair and Comprehensive Evaluation of Routers in Collaborative LLM Systems routing

- **Wanxing Wu**, He Zhu, Yixia Li, ..., Guanhua Chen[†]
- Under Review, EMNLP 2026 [arXiv:2602.11877]
- Built **RouterXBench**, a 3-axis evaluator (AUROC / LPM-MPM-HCR / 6 ID-OOD benchmarks), and used it to show that **internal hidden states are a more discriminative routing signal** than verbose, logit, and external-embedding alternatives, with the finding holding across small models of different families and sizes.

Can Weak Models Explore? Scope Then Cover for LLM Agent Adaptation agentic

- **Wanxing Wu***, He Zhu*, Guanhua Chen[†] (*Equal contribution)
- Under Review, 2026
- Adapting an LLM agent to a new environment hinges on exploration. Two controlled probes surfaced two recurring phenomena: scaling the explorer model brings no gain when the environment is uninformative about target location, and all scales collapse to random search once the candidate space outgrows the per-step inspection budget. Both reveal the same structure inside exploration: scoping a promising subspace is capability-bound, while verifying whether the target lies inside it is not. We propose **scope-then-cover**, where a capable actor nominates a small high-recall subspace and frozen cheap workers cover it in parallel, exposed as a single `explore(scope, query)` action. The same action drops into both a frozen test-time actor and an online experiential learning host, and lifts pass rate over no-explore baselines on ALFWorld, WebShop, and SWE-Bench-Verified.

AutoInterpBench: Level-1 Grounded Benchmarking for Autonomous Mechanistic Interpretability Discovery autoresearch

- Xuanzhe Xu, He Zhu, **Wanxing Wu**, Guanhua Chen[†]
- Under Review, EMNLP 2026
- Built **AutoInterpBench**, a three-level benchmark (264 tasks, 764 GPU runs, 14 models) for *autoresearch* agents on mechanistic interpretability, paired with a **harness-driven discovery** loop that uses Level-1 scores as fitness signals to refine the agent's analysis primitives.

RESEARCH EXPERIENCE

🔥 SUSTech-NLP Group

Research Assistant, advised by Prof. Guanhua Chen.

Shenzhen, China

Jan 2025 – Present

(Co-)first author on the two papers above. For RouterXBench/XRouter, built a vLLM-based multi-model inference and evaluation pipeline covering batch inference, router evaluation, and logit/embedding/hidden-state routing baselines. For Explore-Agent, trained Explore-Agent with OPCD, skill-sd, and Agent-RL; part of this work contributed to the Scope-then-Cover submission.

🏢 Institute of Intelligent Medical Research (IIMR), BGI

Machine Learning Intern (AI for Genomics).

Shenzhen, China

Jul 2024 – Jan 2025

Built a cfDNA data-processing and ML modeling pipeline for early colorectal cancer detection genomics, combining tabular cfDNA features with a BERT-style end-motif sequence encoder; handled feature cleaning, statistical filtering, model-guided feature selection, ensemble modeling, and cross-validation. Reached **94.88% sensitivity** on the held-out cohort.

🏥 Peking University Shenzhen Hospital

Data Analyst Intern.

Shenzhen, China

Jul 2023 – Oct 2023

Clinical Outcome Analysis clinical: Performed biostatistical analysis on multi-timepoint clinical records to compare postoperative outcomes across surgical methods.

HONORS & SKILLS

Honors: Outstanding Graduate, Zhiren College, SUSTech (2025); Outstanding Graduation Thesis, Top 5%, SUSTech (2025); Outstanding Student Scholarship, SUSTech (2024).

LLM & Dev: vLLM, SGLang, agentic-RL workflows; Python, PyTorch, Docker, Linux, Git. **Languages**: English (IELTS

6.5, CET-4/6); Chinese (Native).